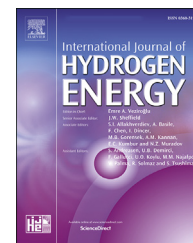


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Experimental hydrogen sorption study on a LaNi₅-based 5 kg reactor with novel conical fins and water tubes and its numerical scale-up through a modular approach

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HIGHLIGHTS

H₂ storage reactor with internal 12 conical fins+4 tubes is experimentally studied. Absorption by 5 kg LaNi₅ under various operating parameters are compared. Atmospheric Desorption for different initial bed temperatures are observed. 5/10 reactors in series requires 2/3 times water flow rate to match performance. Parallel requires proportional water flow rate but facilitates uniform absorption.

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ABSTRACT

Hydrogen sorption characteristics of 5 kg-LaNi₅ reactor featuring conical fins and heat transfer tubes is experimentally studied at various water flow rates (V_s : 2.5, 5, 7 LPM), inlet temperatures (T_s : 30, 25, 20 °C), hydrogen pressures (P_s : 10, 15, 20, 25 bar). Higher V_s , P_s , and lower T_s yield faster absorption through higher driving force ($\propto \ln(P_s/P_{eq})$, P_{eq} : equilibrium pressure). Ninety percent absorption takes ~12.3 and ~13.7 min at V_s : 7 and 2.5 LPM (P_s : 15 bar, T_s : 30 °C). Faster atmospheric desorption needs higher T_s and/or V_s offering higher driving force ($\propto (1-1/P_{eq})$). 90% desorption takes ~20.1 and ~21.8 min for initial bed temperatures of 60 and 30 °C, (V_s : 5 LPM, T_s : 60 °C). Absorption of this 'Single Reactor' and five/ten (25/50 kg) such reactors in series and parallel are studied numerically. Series configuration stores five/ten-times hydrogen than 'Single Reactor' within the same duration by doubling/tripling V_s (same P_s , T_s). Parallel configuration requires proportional water supply.

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Introduction

Hydrogen which can be generated by renewable means [1–3] has the potential to replace conventional fossil fuels for

various applications [4–7]. Hydrogen storage in solid hosts viz. Various metal hydrides (MH) based on LaNi₅ [8–13], FeTi [14–16], Mg [17–20] etc. Has been investigated by several research groups. Two of the main technical challenges of

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hydrogen storage through metal hydrides (MH) are (i) thermal management during absorption/desorption due to the poor thermal conductivity of MH - Hence, bulk MH-bed systems are often embedded with a heat transfer system (HTS) [21,22]; and (ii) efficient and effective scale-up.

Among the major strategies to improve heat transfer, the systems containing heat transfer tubes are widespread. For example, Askri et al. [23] and Kaplan [24] modeled hydrogen absorption in an SS tank with and without water cooling. With the inclusion of a single tube makes the process 56% faster for 90% absorption at hydrogen supply pressure (P_s) of 8 bar [23]. Increasing the overall heat transfer coefficient (U) from 5.5 to $133 \text{ W m}^{-1} \text{ K}^{-1}$ by water jacketing a simple air-cooled SS reactor was reported to boost the absorption performance [24]. The effect of multiple straight heat transfer tubes through the MH bed was extensively investigated by Muthukumar et al. [25]. They found that increasing the number of cooling tubes from 12 to 20 makes the heat transfer and absorption faster. Anbarasu et al. [26–28] experimentally compared the absorption process of 36 and 60 cooling tube systems with 2.75 kg $\text{LaNi}_{4.91}\text{Sn}_{0.15}$ (Lm: La based misch metal). For a given inlet water supply velocity (v_s : 0.33 m s^{-1}) and temperature (T_s : 30°C) the absorption was 25% faster for the 60-tube system at P_s : 25 bar [26] and for the same v_s and T_s : 50°C desorption became 44% faster [27]. Raju et al. [29] provided a comprehensive set of arithmetic equations to design high capacity reactors featuring straight tubes. Kumar et al. [30] demonstrated a large reactor using this design approach with 99 tubes and 40 kg of $\text{LaNi}_{4.7}\text{Al}_{0.3}$. Other studies on shell and tube type reactors were also reported by Bao et al. [31] and Mohan et al. [32] for different storage capacities. Gkanas et al. [33] proposed a tank with rectangular cross-section instead of the conventional cylindrical tank. Their simulations showed the effect of bed dimensions and tube arrangement. The design proposed by Wang et al. [34,35] features branching tubes. These branch-like tubes originated from tubes inside the reactor and thus increasing the effective heat transfer surface significantly [34,35]. As a result, more than 55% improvement in absorption time was obtained from their optimized design [35]. Jana et al. [36] compared the performance of a 40-cooling tube design with a 19-tube design containing $\text{MmNi}_{4.6}\text{Al}_{0.4}$ which possesses a common cooling shell. They showed almost similar performance for the same sorption conditions. However, the latter system has 17% better MH-to-container weight ratio than the former [36]. A similar designs was also proposed by Krokos et al. [37] featuring several tubes filled with MH inside a common water jacket. They studied the effect of five different arrangements of these MH tubes. Apart from the straight tubes, the helical tubes and those with elliptical cross section have also been used in LaNi_5 -based reactors. For example, Mellouli et al. [38] showed that a single helical SS tube (T_s : 20°C) reduces the time for 80% absorption by 1 kg LaNi_5 (P_s : 10 bar) from 5000 to 1000 s. In another study, Tiwari et al. [39,40] optimized the various design parameters of a Mg_2Ni -based hydrogen storage system with a helical tube. Another optimization study on helical tube with a central return tube by Eisapour et al. [41] suggested the selection of heat exchanger tube diameter, coil diameter and the coil pitch as 5, 18 and 10 mm, respectively, for an optimized design of an Mg_2Ni reactor with 50 mm diameter and 80 mm height [41]. Wang

et al. proposed a novel cooling system using duplex helical tubes with elliptical cross-section [42] to achieve a higher contact area between the tube and the LaNi_5 bed than that with a circular cross section tube.

Several studies combined the effect of heat transfer tubes and fins to achieve the advantages of the both. Dhaou et al. [43] improved the design of Mellouli et al. [38] by adding a helical copper fin to halve the time required for 75% absorption by 1 kg LaNi_5 . In another study, Mellouli et al. [44] showed that for a 1 kg LaNi_5 reactor the time taken for 90% absorption decreased by three times to 500 s when two helical tubes are used against one at P_s : 10 bar. They demonstrated that the same improvement in the performance can also be achieved by combining the SS tube with a helical copper fin. In their study on 1 kg LaNi_5 reactor, Singh et al. [45] used flat copper fins with punched holes together with a central cooling tube. The unperforated fins were 44% heavier and reduced the absorption (1 wt%) time by 13%. Lewis et al. [46] showed that a reactor with 4 U-tubes (T_s : 20°C , v_s : 0.3 m s^{-1}) could achieve 90% absorption within 998 s (P_s : 10 atm). As expected, this duration reduced to 867 and 656 s with the attachment of one and three fins, respectively, to each of the U-tubes. Keshari et al. [47] employed straight, circular and spiral copper pin-fin configurations welded with copper tubes for a reactor with 1 kg LaNi_5 capacity. They reported that the centre-to-centre pin-fin configuration showed a better sorption performance than the side-by-side configuration [48]. A Bio-inspired leaf-vein type fin configuration was explored by Krishna et al. [49]. They compared such two configurations viz. with central straight tube and narrow trapezoidal channels with longitudinal fin arrangement for a 10 kg LaNi_5 reactor. An optimized trapezoidal channel design was reported to have 73% better performance than the longitudinal fin arrangement. The poor thermal conductivity of the MH material is the main constraint in the way of effective thermal management. Hence, strategies such as reducing the thickness of the MH bed region in longitudinal and radial direction to lower the thermal resistance of the MH should be employed as shown in our previous numerical study [50]. Designs involving embedded cooling tubes can take advantage of similar modification by implementing a large number of tubes closely placed together [25–29].

From the studies summarized above, it is evident that there are multiple ways to improve performance of a hydrogen storage reactor. However, most of the above mentioned experimental and numerical studies featuring the fin and/or tube HTS focus on single reactors of 1 kg or less amount of metal hydride. However, a practical hydrogen storage reactor should possess high gravimetric capacity for a given application. This makes the efficient and effective scale-up of the hydrogen storage system as one of the most important aspects of research.

In our systematic numerical studies [50], we demonstrated that a simple proportional scaling-up of a reactor of any size does not yield a better performance than that from a smaller reactor even if the heat transfer fluid flow is increased proportional to the increased capacity. We explored the effect of various design parameters viz. fin/MH-bed and fin/tube contact areas; tank and cooling tube diameters; and the number of tubes on the performance of a single small reactor (5 kg LaNi_5). Among these design parameters, the fin/MH-bed

contact area and the tank diameter are the most effective. Using these parameters, we designed an optimized a single scaled-up reactor with 25 kg of LaNi_5 . As a part of the scale-up effort, we also explored the series and parallel configurations of five small reactors (each with 5 kg of LaNi_5) through modular approach to match the performance of the optimized scaled-up single reactor [50]. We found that for the same total water flow rate, the series configuration performs better than the parallel counterpart. Similar modular approaches for scaling-up were also proposed by other groups. For example, Chung et al. [51] proposed to use a common water bath to supply/extract heat to/from reactors featuring heat pipes. Though their design does not need any water flow, the cooling/heating has to be proportionally increased with the number of the reactors. A design of stackable MH-containers with heat exchanger plates carrying fluid between two such plates was proposed by Chippar et al. numerically [52]. In the studies of Jana et al. [36] with 25 kg $\text{MmNi}_{4.6}\text{Al}_{0.4}$, they suggested that four such reactors can be used together. These designs require a proportional increase in the water flow rate [36,52]. All these proposals suggested parallel configuration of the MH-reactors to take advantage of the same water inlet temperature for all the reactors and minimal pressure drop. Such a modular approach can offer ease of maintenance and flexibility in operation and configuration.

In this work firstly, experimental studies on the performance of a lab-scale 5 kg LaNi_5 reactor ('Single Reactor') with internal HTS featuring fins and tubes are conducted. Our earlier numerical study [53] showed that the conical fins of this internal HTS can improve the performance over a flat circular fin design by providing a greater surface area without hampering the hydrogen flow through the MH-bed. Here, various V_s (2.5, 5, 7 LPM), P_s (10, 15, 20, 25 bar), and T_s (30, 25, 20 °C) values are employed for studying the sorption performance of the single reactor and the results are analyzed from the sorption-driving force point of view. Eventually, the performance of this single reactor is numerically modeled and validated. Finally, the performances of the scaled-up configurations of five (25 kg LaNi_5) and ten (50 kg LaNi_5) such single reactor modules connected in series and parallel are evaluated numerically. The choice of a 5 kg reactor can facilitate easier scale-up to either 25 or 50 kg than a 1 kg reactor – which is used by several research groups. The series configurations of such 5 kg reactors can store five- and ten-times hydrogen than the single reactor within the same duration by just doubling and tripling the water flow rate, respectively, at the same P_s and T_s . However, the parallel configuration has an advantage of being supplied with water at the same temperature to all the reactors. The present work shows the comparison of the performances of series and parallel configurations.

Experimental procedure

The present work consists of two parts viz. (i) experimental hydrogen sorption in 5 kg LaNi_5 reactor with the novel conical fins and tubes, and (ii) numerical studies on the scale-up using such 5 kg reactors in a modular way. The choice of LaNi_5 as the MH is due to its capability to absorb hydrogen at 30 °C (i.e. room temperature in author's region) and at moderately low

pressures. Also, this material is used in several studies by other researchers which enables the comparison between these studies and the present one.

Reactor description

The design and the photograph of the reactor are shown in Fig. 1a. An SS 316 tank made of the commercially available standard tubes of an outer diameter (OD) of 88.9 mm and a thickness of 5.49 mm (Sch- 40s) is used to fabricate the reactor. The length of the reactor is 340 mm. The reactor is closed at both ends by two SS 316 flanges. The design and the photograph of the internal heat transfer system (HTS) comprising of the conical fins and the tubes is shown Fig. 1b. From Fig. 1b, the HTS consists of 12 conical copper fins with a semi-cone angle of 45°. These fins are coaxially connected with a pitch of 28 mm by brazing with four SS 316 tubes of 1/4th inch OD. The axes of these tubes is parallel to that of the cones. The tubes are arranged around the axis of the fins at a sectoral angle of 90°. The height of the conical fins is 28.5 mm. The thickness and the inner diameter (ID) of the fins are 1 and 16 mm, respectively. Along the axis of these fins a central sintered SS 316 filter (OD = 12 mm) is arranged. This filter acts as a hydrogen artery for ensuring a uniform distribution of gas along the height of the reactor. There is a clearance of 2.5 mm between the OD of the fins and the inner wall of the tank. Similarly, a clearance of 2 mm is provided between the ID of the fins and the OD of the filter. Three K-type thermocouples are inserted perpendicularly into the reactor at an equal separation along its length (Fig. 1a). These thermocouples are aligned along a straight line when seen from the outside of the reactor. The thermocouples protrude into the reactor up to almost midway within the gap between the inner wall of the tank and the OD of the filter. The sectoral angle made by the tip of the thermocouples with the tubes is 45°, with respect to the axis of the reactor.

LaNi_5 powder is loaded through one end of the tank while keeping the other end closed. The conical design of the fins and the inner and outer clearances are advantageous because of the funneling effect they provide which aid in LaNi_5 powder sliding easily over the fins to fill the reactor. The reactor is sealed by bolting the flange in its place after filling with the powder. A Teflon gasket is placed between the flange and the tank-end before sealing the reactor to prevent any hydrogen leaks.

Experimental set-up and leak test

The experimental set-up consisting of hydrogen and argon cylinders, Coriolis mass flow meter (Emerson CMFS015 M), pressure transducer, reactor, thermostatic bath-cum-circulator, rotameter, vacuum pump, various valves and data logger is shown in Fig. 2. The gas flows through a 1/4th inch SS 316 tubing. Water is used as the heat transfer fluid (HTF) and is circulated by the circulator that can deliver up to 7 L per minute (LPM). To adjust the water flow to various desirable rates, an arrangement of 2 valves and a rotameter is utilized (Fig. 2). The thermostatic water bath (5L) consists of a cooling (1 kW) and also a heating coil (2 kW) can keep the water at a pre-set temperature. The water flow is channeled to the HTS inside the

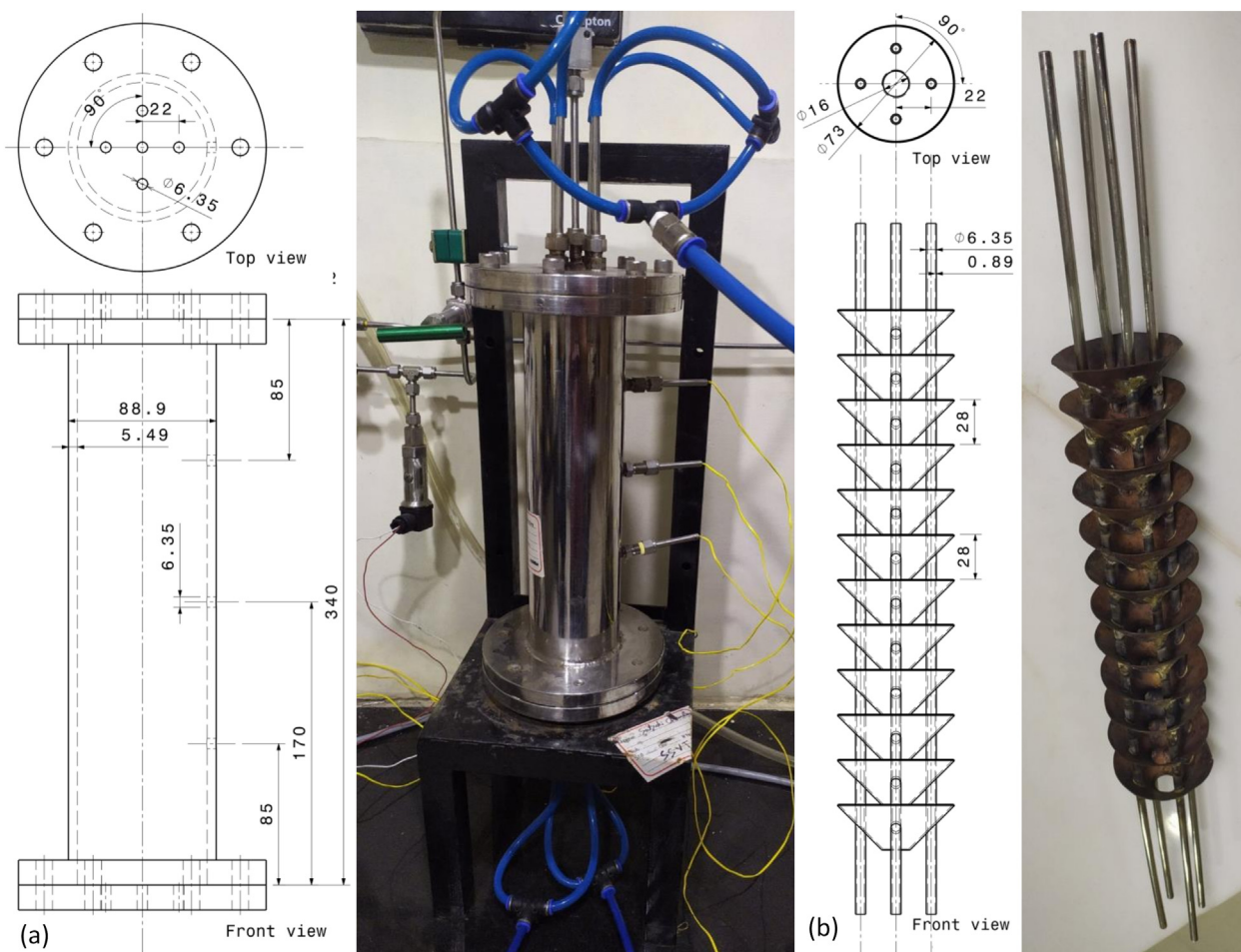


Fig. 1 – 5 kg LaNi₅ reactor with novel conical fins and cooling tubes: (a) reactor tank, (b) fin + tube system.

reactor through PU tubing which can endure up to ~ 60 °C. Various data generated throughout the process is recorded using the data logger at a scan rate of 1 Hz. The temperature data from the K-type thermocouples (uncertainty ± 1 °C) is recorded with a least count of 0.1 °C. The mass flow meter requires a 230 V AC power supply and generates a current signal within 4–20 mA, which can be set to denote a defined range of mass flow rates employing ProlinkIII software. The pressure transducer requires a 24 V DC power supply to give an output signal within 4–20 mA, which is linearly proportional to 0–100 bar absolute pressure. The data logger reads these current signals with a least count of 0.01 mA and have an uncertainty of $\pm 0.01\%$ of the full scale deflection for temperature as well as current value.

Gas leakage from the various components mentioned above can occur through any possible cracks within the welded zones and other junctions. For preliminary leak testing, the empty reactor tank was pressurized with argon to 100 bar and soap-water was applied at various areas which are prone to leakages. Following the preliminary leak test, the pressurized reactor was left for 24 h to check for any significant drop in argon pressure. The same procedure is also used for leak testing with hydrogen gas. After a thorough observation, necessary repairs were made at various locations and any suspected gaskets etc. are replaced. Water leakage is also possible through any cracks that can be present in the HTS. Hence, water is

pumped through HTS for 30 min prior to filling the reactor with LaNi₅ to check for the possible water leakage inside the reactor. After assuring the absence of the gas and water leakage the reactor is filled with LaNi₅. Eventually, the aforementioned gas leakage tests were repeated on the filled reactor assembly.

Activation and hydrogen sorption by LaNi₅

The LaNi₅-filled reactor was evacuated while heating to 55 °C for removing air and any residual gas. Eventually, the filled tank is purged with argon while heating followed by another evacuation step to ensure the removal of any further residual gases and moisture. For activation, LaNi₅ bed is exposed to 15 consecutive absorption and desorption cycles using hydrogen (99.999% pure) until the absorption capacity approached the maximum. After every desorption, the reactor is purged with argon. The absorption steps are performed at P_s : 15 bar and T_s : 5 °C. The desorption steps are conducted at T_s : 55 °C while evacuating. Recent studies by Patel et al. [54] show that MH can be activated inside a pressurized milling vessel by shaking the whole assembly. However, the sample size mentioned in their work is around 20–100 g, which helped in improving hydrogen sorption behavior tremendously. Since the weight of LaNi₅ used in present study is 5 kg, it is necessary to use suitably large milling vessel for activation. Hence, such an intensive milling may not be appropriate for the present study.

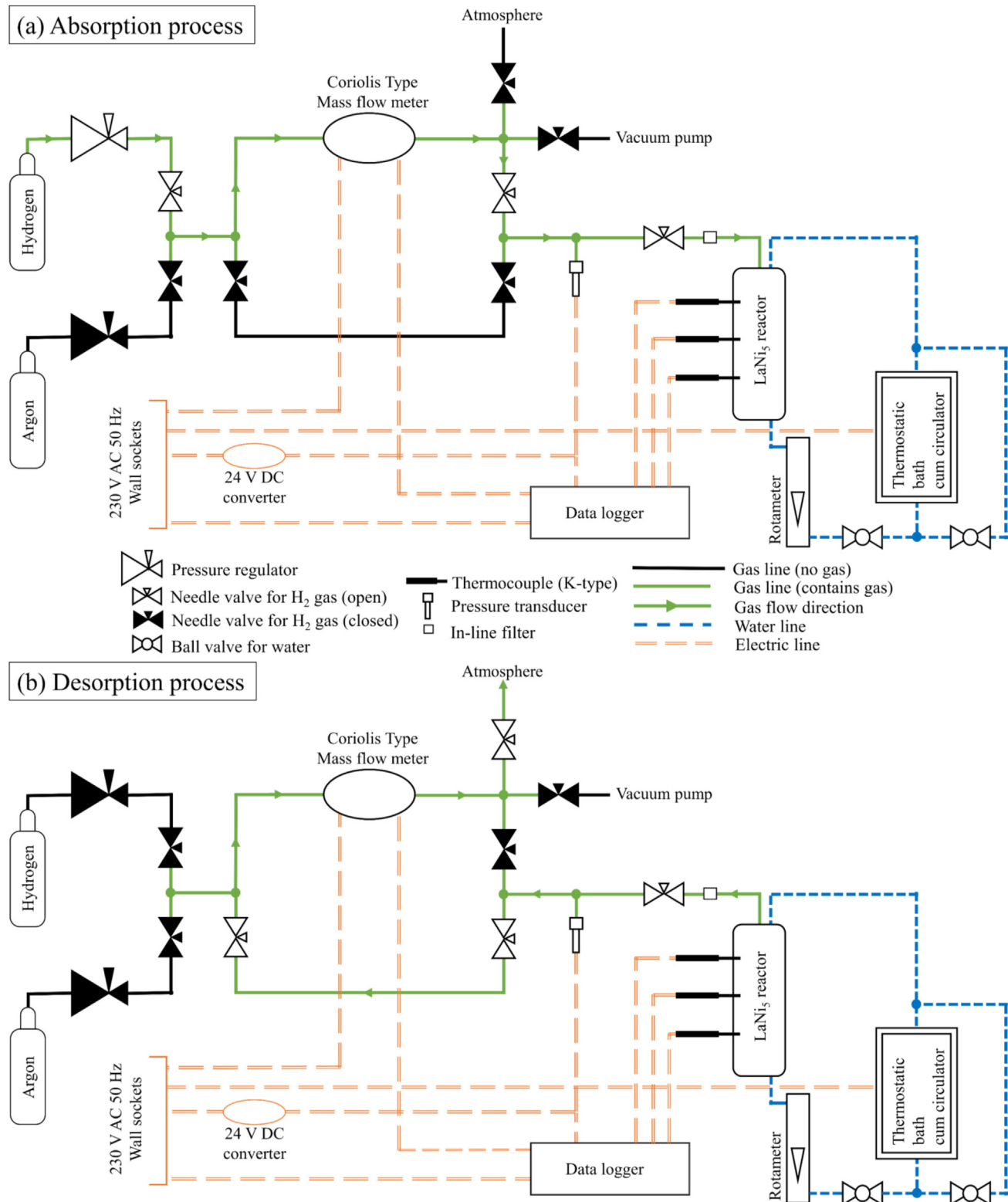


Fig. 2 – Experimental setup and gas flow direction for (a) absorption and (b) desorption processes.

The actual hydrogen absorption and desorption are described schematically in Fig. 2a and b, respectively. For continuous exothermic absorption and the endothermic desorption, cooling and heating of the LaNi₅ (hydride) is required, respectively. A continuous flow of cold (T_s : 20, 25 and

30 °C) or hot water (T_s : 50 and 60 °C) is employed to achieve the aforementioned cooling or heating. The performance parameters are (i) the LaNi₅ (hydride) bed temperature measured by the thermocouples, and (ii) the hydrogen flow into or out of the reactor as measured by the mass flow meter.

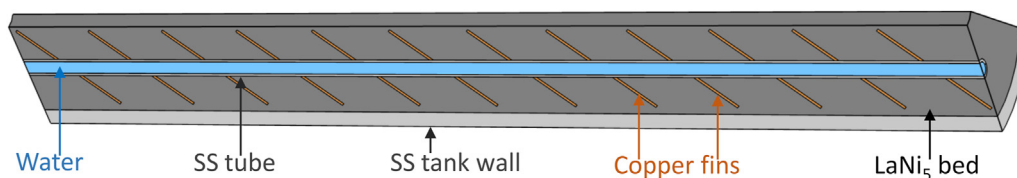


Fig. 3 – 1/8th longitudinal section of the reactor showing various simulation domains viz. water, SS tube, SS tank wall, Copper fins, LaNi₅ bed.

Numerical procedure

In our earlier reports [50,53], we discussed about the performance of a 5 kg LaNi₅ based hydrogen storage reactor [53] and its scale-up to 25 kg [50]. Among the various scaled-up designs discussed, the designs having five 5 kg reactors connected in series or parallel through water-line in a modular way resulted in promising performance [50]. Therefore, this modular approach is used here employing the design of the experimental reactor described above as a single module. The performances of the modular configurations consisting of five and ten such 5 kg reactors connected in series and parallel are studied here numerically.

For performing numerical simulations, hydrogen absorption in the above-mentioned configurations is studied by employing COMSOL 5.4 which uses finite element method. A 1/8th longitudinal section of the 5 kg reactor is used to achieve lower computational load. The longitudinal section of the reactor which includes the various computational domains is shown in Fig. 3.

The assumptions made for the numerical study are [23,25,44,45,50,53,55,56].

- Hydrogen behaves as an ideal gas due to the fact that the pressure and temperature considered for the present study are well above their critical values.
- Local thermal equilibrium between the immobile porous LaNi₅ bed and the hydrogen gas is valid.
- The physical and thermal properties of LaNi₅ are constant within the chosen pressure, temperature and the amount of the absorbed hydrogen.
- The porosity and volume of LaNi₅ bed do not vary during any sorption process.
- Heat transfer by only conduction is considered with an effective conductivity.

The mathematical equations and the initial and boundary conditions of the numerical scheme along with the properties of LaNi₅ are mentioned in our earlier reports [50,53] and in the Supplementary Information S1 and S2.

Results and discussion

Hydrogen absorption and desorption by 5 kg reactor

The hydrogen absorption and desorption experiments of the fabricated 5 kg reactor were conducted at various values of hydrogen pressure, inlet water temperature and its flow rate.

Prior to discussing the results, a practical limitation of the Coriolis mass flow meter employed for measuring hydrogen flow rates into and out of the reactor is described here. The errors associated with hydrogen flow rate are shown in Fig. S1 in Supplementary Information S3. From Fig. S1, a decrease in the hydrogen flow rate results in a drop in the accuracy of the mass flow meter. For example, at a hydrogen flow rate of 25 g min⁻¹ the error in the mass flow meter reading is within ~0.54%; whereas, at 1.5 and 1 g min⁻¹ the errors are within 10% and 13.56%, respectively (Fig. S1). Also, the steep rise in the error at hydrogen flow rates lower than 0.5 g min⁻¹ can be observed from Fig. S1. As per the manufacturer's suggestion, a certain positive value should be set as a cut-off. In our system, when all the valves of the gas circuit are closed leading to a no-flow situation, the meter erroneously detects a finite flow reading when the cut-off value is set below 1.5 g min⁻¹. Hence, 1.5 g min⁻¹ is chosen as the cut-off for the present study. Here, the recorded hydrogen flow rates are also presented for the low flow region. However, for estimation of the total mass of hydrogen absorbed or desorbed the flow rate curves are integrated and reported up to only 90% of the maximum gravimetric capacity of LaNi₅ i.e. 63 g (1.26 wt % as against the maximum hydrogen capacity of 1.4 wt%) for 5 kg of material. There are around ±5% error associated with the reported absorption cases for 63 g of hydrogen and that value is ±6% for the reported desorption cases.

Absorption performance

The absorption performance can be understood from the point of view of its (i) driving force, which is proportional to $\ln(P_s/P_{eq})$ at any given temperature and (ii) resistance for absorption, which increases with increased hydrogenation extent (i.e. $\rho'_{sat} - \rho'_{MH}$). Faster hydrogen absorption rate could be attained by higher magnitudes of $\ln(P_s/P_{eq})$ and $\rho'_{sat} - \rho'_{MH}$. Various V_s , P_s , and T_s can influence these terms. For the present study, the hydrogen absorption characteristics of the reactor containing 5 kg of LaNi₅ are studied by varying only (i) water flow rate (V_s : 2.5, 5 and 7 LPM), (ii) supply pressure (P_s : 10, 15, 20 and 25 LPM), and (iii) temperature (T_s : 20, 25 and 30 °C). The effects of these parameters on the experimental performance of the 5 kg reactor are shown in Fig. 4.

Effect of HTF flow rate: For studying the effect of HTF (water) flow rate, three different flow rate values viz. 2.5, 5 and 7 LPM were chosen while keeping P_s : 15 bar and T_s : 30 °C. Here, T_s : 30 °C is chosen as it is close to the ambient temperature in the authors' region. The hydrogen flow rate into the reactor, average temperature, and total hydrogen absorption are shown as functions of time in Fig. 4a, b, and c, respectively. The hydrogen flow rate into the reactor is almost the same for all the chosen water flow rate values up to ~2 min and shows

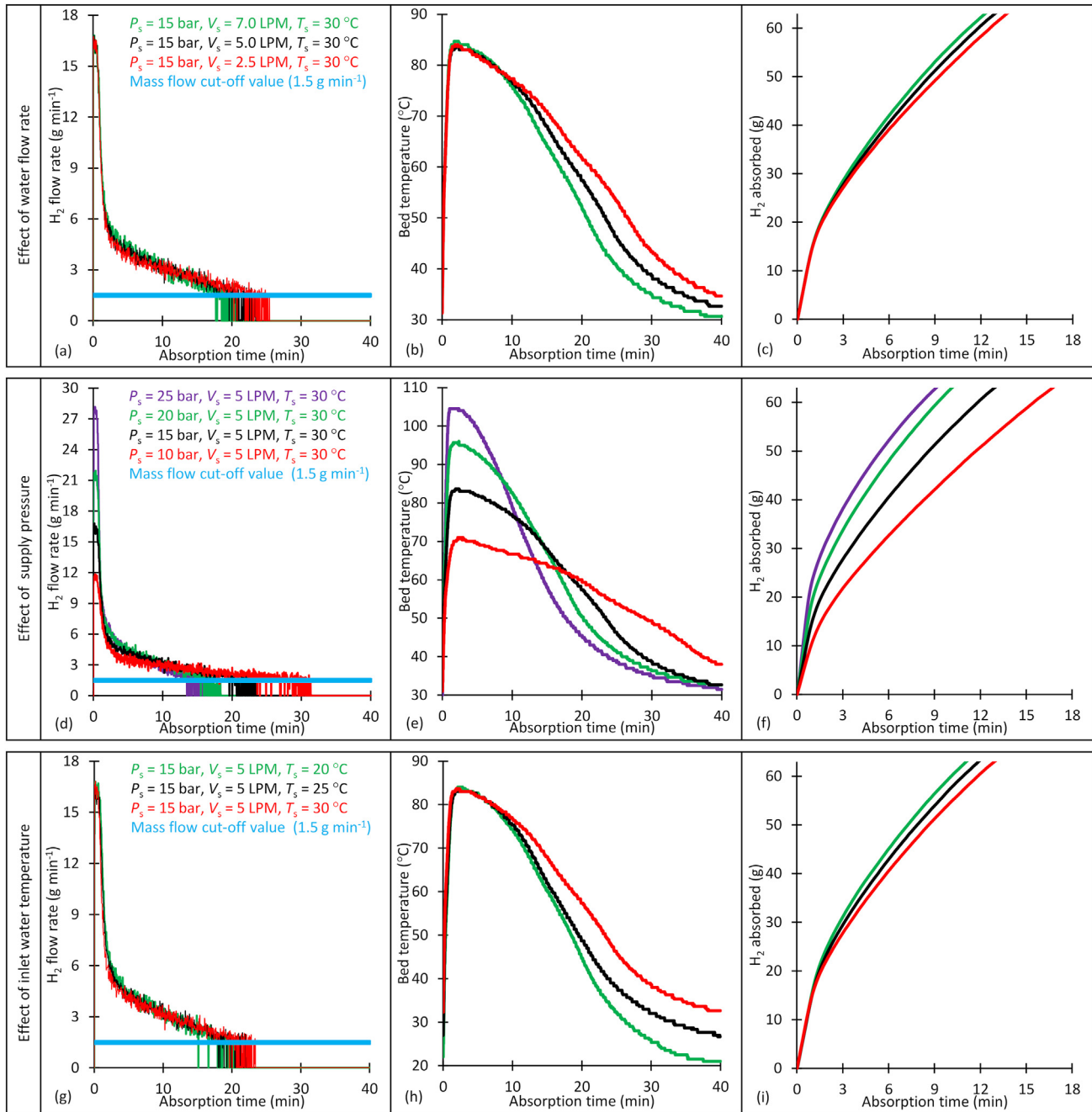


Fig. 4 – Effect of various operational parameters viz. (a–c) water volume flow rate: V_s , (d–f) hydrogen supply pressure: P_s , and (g–i) water inlet temperature: T_s on absorption process.

slight variation beyond that (Fig. 4a). This is also reflected in the changes within the average temperature (Fig. 4b) and the total H_2 absorbed (Fig. 4c) up to this time. The steep rise in the bed temperature (Fig. 4b) is due to the heat generated at a high rate in the reactor resulting from the rapid exothermic hydrogen absorption by $LaNi_5$ as seen from Fig. 4c. The rapid temperature rise dominates the effect of cooling offered by the water for the chosen flow rate values resulting in the overlapping curves (Fig. 4a, b, and c) up to ~2 min.

The domination of water cooling is evident beyond ~2 min as shown in Fig. 4b, leading to the deviation of total H_2

absorbed (Fig. 4c). As expected beyond ~2 min, the dominant water cooling leads to the rapid temperature drop (Fig. 4b) and results in the highest total absorbed H_2 (Fig. 4c) at any given time for V_s : 7 LPM. This highest volume flow rate of water leads to the lowest temperature at any time (beyond ~2 min). This lowest temperature, in turn, leads to the lowest P_{eq} . Since the P_s is the same in all the cases here, the driving force ($\propto \ln(P_s/P_{eq})$) is the highest for V_s : 7 LPM. Within the initial ~2 min, however, the almost same temperature in all the different water flow rate cases results in the same equilibrium pressure. This leads to an almost similar magnitude of the driving

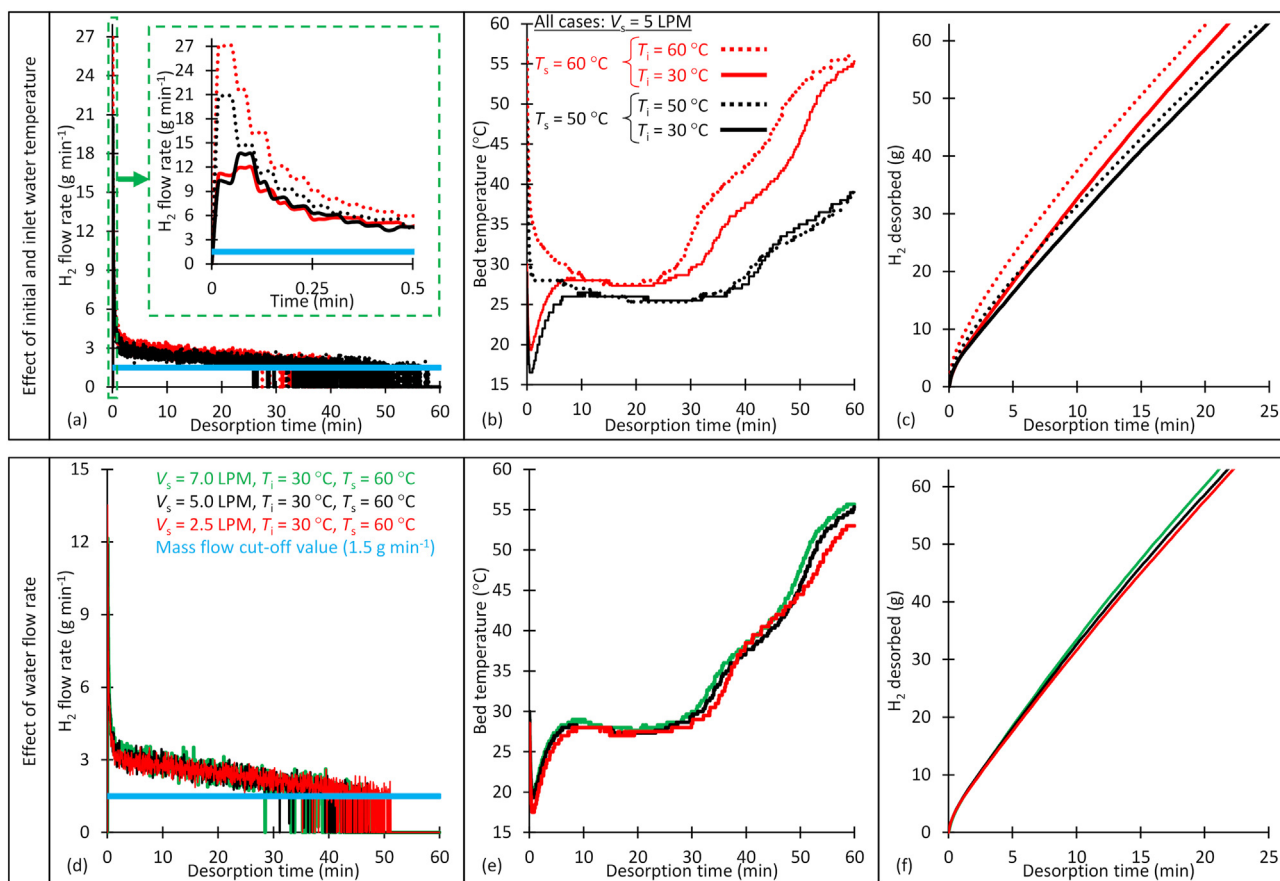


Fig. 5 – Effect of various operational parameter viz. (a–c) reactor initial temperature: T_s , water inlet temperature: T_i , and (d–f) volume flow rate: V_s on desorption to ~1 atm pressure.

force and thus the similar extents of hydrogen absorption (Fig. 4c). For absorbing 90% of the maximum capacity of LaNi₅ (i.e. 63 g hydrogen), the system takes ~12.3 and ~13.7 min at V_s : 7 and 2.5 LPM, respectively.

Effect of supply pressure: Four values of P_s are considered viz. 10, 15, 20 and 25 bar of hydrogen to study its effect keeping the V_s and T_s constant. The obtained results are plotted in Fig. 4d–f. The total hydrogen absorption is different in all the cases considered right from the beginning. As expected, the highest P_s results in the highest extent of hydrogen absorption (Fig. 4f). Accordingly, the H_2 flow rate drops rapidly (Fig. 4d) and the average temperature rises rapidly in this case (Fig. 4e) during the initial stages. Eventually, the absorption rate decreases due to the increased resistance for further absorption which is manifested as $\rho'_{sat} - \rho'_{MH}$. The highest extent of hydrogen absorption in P_s : 25 bar leads to an attainment of 90% (63 g) maximum capacity in this case within just ~9.1 min of the process (Fig. 4f). Also, an almost 100% absorption occurs by ~14 min (not shown in Fig. 4f). Accordingly, the H_2 flow rate into the reactor drops to negligible values after this time (Fig. 4d). Beyond 14 min, the effect of water cooling starts to dominate in removing the residual heat from the system leading to a rapid drop in the temperature (Fig. 4e). This domination of water cooling in removing the residual heat can also be observed in the cases of other P_s values (Fig. 4e). After ~14 min, there is still some heat generation in the reactor at P_s :

10 and 15 bar for a certain time. This is because 90% absorption occurs at ~16.7 and ~12.9 min at these respective P_s values. Hence, water cooling is not yet dominant in these cases. This manifests as a reverse trend in the temperature curves beyond ~14 min. The reverse trend in the temperature curves is also a result of the total heat removal being almost constant in all the cases.

Effect of inlet temperature of HTF: The effect of the inlet temperature of the HTF is studied by considering three different water inlet temperatures viz. T_s : 20, 25 and 30 °C. The colder water results in faster cooling as seen from Fig. 4h. The P_{eq} is lower when T_s is lower. Hence, the driving force for hydrogen absorption ($\propto \ln(P_s/P_{eq})$) is expected to be higher in this case (i.e. at lower T_s), resulting in faster absorption (Fig. 4i). It is interesting to observe that T_s : 30 °C case takes only ~2 min more than the T_s : 20 °C case to reach 90% absorption (~12.9 min at T_s : 30 °C versus ~11 min at T_s : 20 °C).

Desorption performance

The desorption process was studied at a desorption pressure, $P_d \approx 1$ bar, by considering different water supply temperatures (T_s : 50 and 60 °C), initial bed temperatures (T_i : 30 °C and $T_i = T_s$) and water flow rates (V_s : 2.5, 5 and 7 LPM). At $T_i \approx 60$, 50 and 30 °C the equilibrium pressures for desorption are ~6, ~4.2 and ~2 bar, respectively (Supplementary Information S1). Hence, the driving force ($\propto (1 - P_d/P_{eq})$) decreases in the above

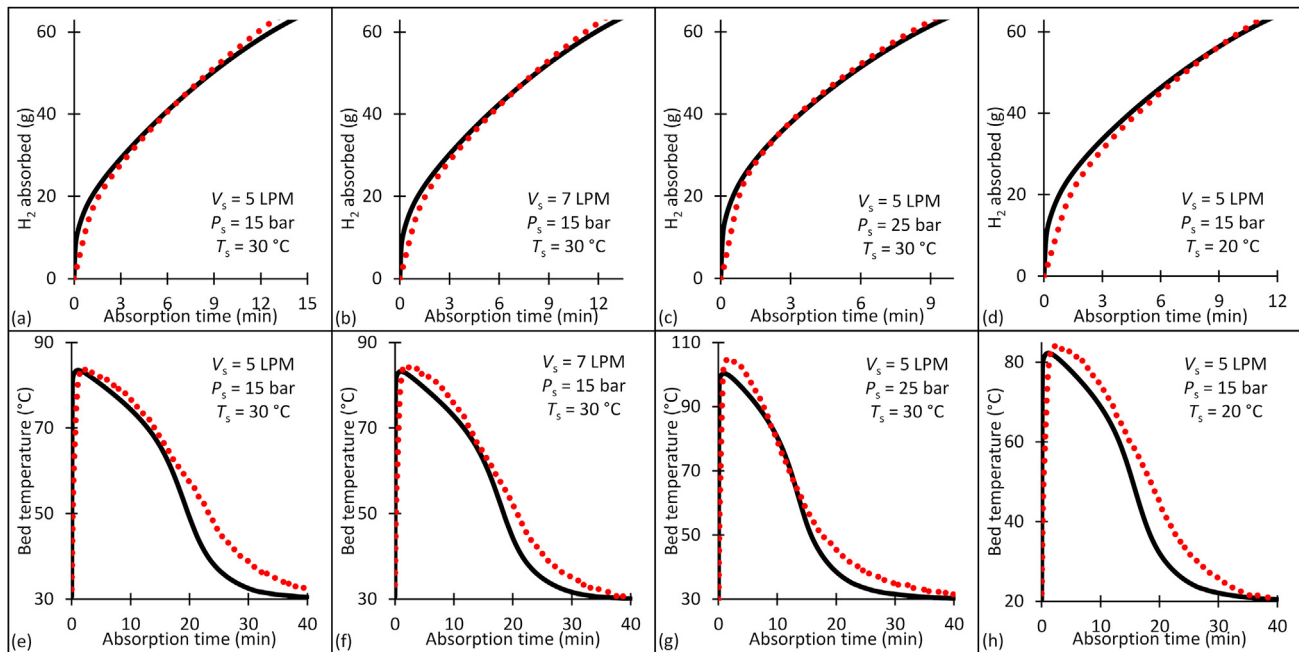


Fig. 6 – Numerical validation of some reported absorption cases with different conditions viz. V_s , P_s , and T_s (Experimental results: dotted red lines, Numerical predictions: solid black lines). (For interpretation of the references to color/colour in this figure legend, the reader is referred to the Web version of this article.)

order of T_i . This results in different initial hydrogen desorption rates viz. ~ 27 , ~ 21 and ~ 14 g min $^{-1}$, respectively (inset, Fig. 5a). Initially, the hydrogen flow rate is high due to steep desorption and drops rapidly (Fig. 5a and d). The steep desorption leads to a fast drop in the average bed temperature (Fig. 5b and e). This is because of the endothermic nature of desorption.

In the cases of T_i : 60 and 50 °C, the temperature dropped to 28 and 25 °C, respectively and the LaNi $_5$ bed stays at that temperature until the desorption process is completed (Fig. 5b). For a certain time, the bed temperature remains constant at these values because the endothermic desorption process is utilizing all the heat supplied by the hot water. In the cases for T_i : 30 °C (T_s : 60 and 50 °C), the bed temperature initially drops below 20 °C and eventually increases up to around the same values of 28 and 25 °C (for T_s : 60 and 50 °C, respectively). Similar to the cases of T_i : 60 and 50 °C, the bed remains constant here for a certain time until the desorption is over. With the temperature drop in the initial stages, the desorption equilibrium pressure also drops to certain value where it almost equals that of the atmosphere, momentarily. This momentary near-equalization of these pressures hinders hydrogen desorption in this duration. This is reflected in the shallow levels of hydrogen flow rate out of the reactor (Fig. 5a and d) and a decrease in the rate of the total hydrogen released (Fig. 5c and f). The momentary hindrance in desorption triggers the rise in the average bed temperature in the case of T_i : 30 °C (Fig. 5b and e). However, this feature of rising temperature cannot be seen in the cases of T_i : 60 and 50 °C because the bed is already at such higher temperatures. With this temperature rise, in the case of T_i : 30 °C, the equilibrium pressure for desorption increases leading to an increase in the driving force ($\propto (1 - P_d/P_{eq})$, Supplementary Information S1) for

desorption. Hence, desorption proceeds to completion within the plateau (Fig. 5b and e). Following the complete desorption, the heat transfer from the water to the bed leads to only increase in the temperature of the latter. This shows that the flow rates chosen for this study do not influence the rise in the average bed temperature, significantly.

The studies show that 90% (i.e. 63 g) of hydrogen desorption of hydrogen for the cases of inlet temperatures of T_i : 60 and 30 °C at a supply temperature of T_s : 60 °C happens within ~ 20.1 and ~ 21.8 min, respectively. The T_i : 60 °C case has a greater initial heat content than the T_i : 30 °C case. Hence higher desorption rate is attained in the T_i : 60 °C case (Fig. 5c) before the bed temperature reaches the plateau (Fig. 5b). A similar argument can be offered in the cases of T_i : 30 and 50 °C at T_s : 50 °C. From Fig. 5c, the gains in the times for 90% desorption at T_i : 60 and 50 °C are just ~ 1.7 and ~ 0.7 min over T_i : 30 °C.

Validation of the numerical simulations

The performance of the reactor with 5 kg LaNi $_5$ is numerically simulated. The grid setting is selected after performing a grid independency test on the 1/8th longitudinal section shown in Fig. 3 (and Supplementary Information S4). The results are validated with the experimental counterparts for absorption. The experimental and simulated hydrogen absorption curves and the temperature profiles of the LaNi $_5$ bed for various scenarios are shown in Fig. 6 to capture the effect of V_s (Fig. 5a and b and Fig. 5e and f); P_s (Fig. 5a, c and Fig. 5e and g) and T_s (Fig. 5a, d and Fig. 5e and h). In general, the numerical results are in good agreement with the experimental counterparts. Any variations between the simulated and experimental absorption curves are due to the following reasons:

- i. The model assumes constant thermo-physical properties, volume and porosity of LaNi_5 bed throughout the sorption process. However, they may vary in the experiments.
- ii. To estimate the total mass of hydrogen absorbed, integration is performed over the mass flow rate with time. This mass flow rate is measured at an upstream point of the gas circuit (Fig. 2a). It takes certain finite time for hydrogen to fill the entire bed in the reactor. However, in the numerical simulation, the control volume is considered to be completely filled with hydrogen in the initial stages. This difference is reflected in the initial stages of absorption (Fig. 6).
- iii. The experimentally measured temperature is an arithmetic average of the readings from three thermocouples positioned at different locations in the reactor. However, in the simulations it represents the volumetric average (Fig. 7).

Numerical scale-up through modular approach

In our previous work, we reported an extensive treatise on the strategies to scale-up the reactors [50]. We demonstrated that the LaNi_5 -bed thickness, fin/tube and fin/MH-bed contact areas play a crucial role in the scale-up without compromise in the performance of the 5 kg reactor system. We also demonstrated that the modular designs consisting of five 5 kg reactors connected in series or parallel with water line can match the performance of the optimized single 25 kg LaNi_5 reactor [50]. Due to the advantages offered by the modular approach in operation, maintenance and configuration, this approach is further exploited here for investigating the efficient ways of scaling up.

For the present study we evaluate the performance of five and ten 5 kg LaNi_5 reactors, corresponding to 25 and 50 kg of LaNi_5 , connected in series and parallel at V_s : 5 LPM, P_s : 15 bar and T_s : 30 °C. These designs are named as ‘5-

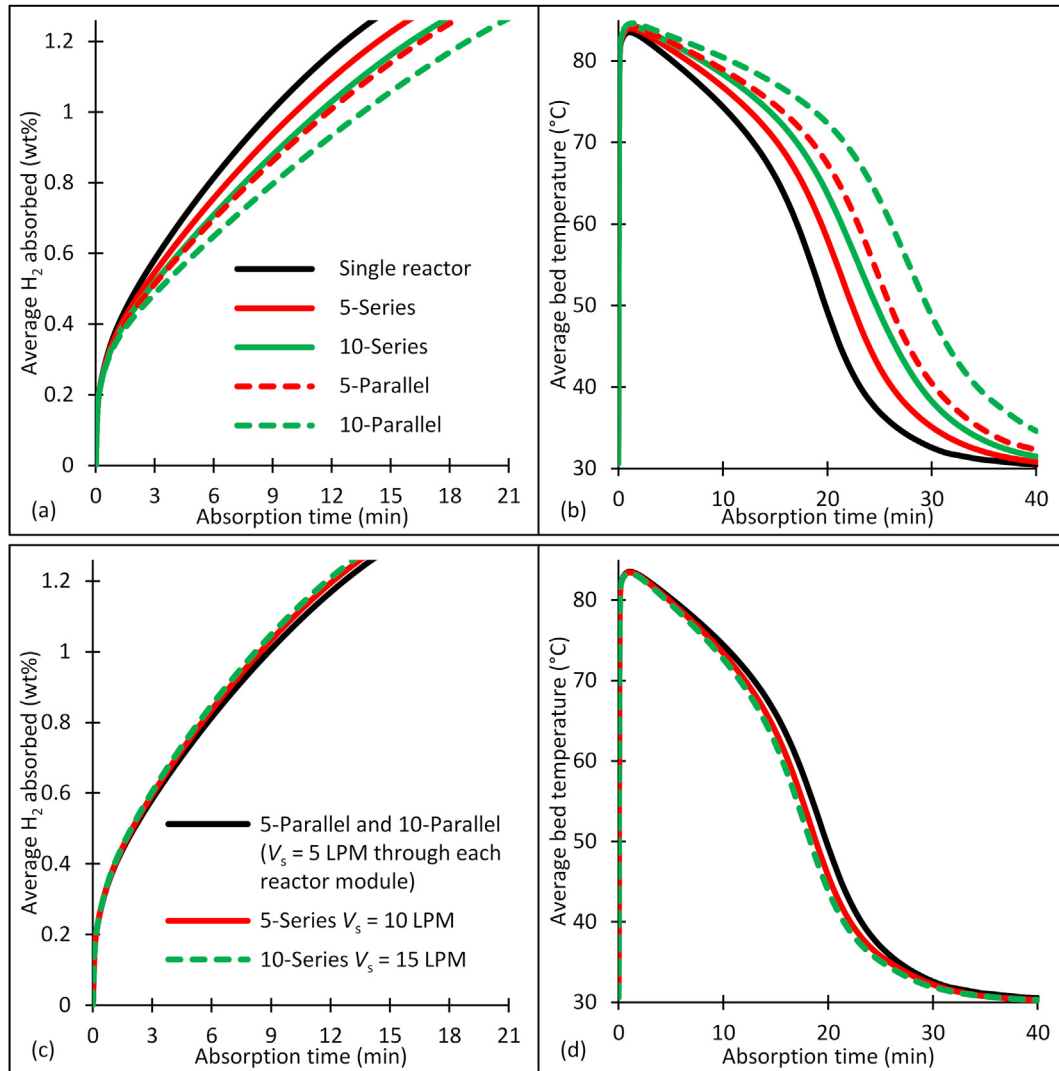


Fig. 7 – Performance of modular designs (a,b) for same external absorption conditions, and (b,d) for $V_s = 10$ and 15 LPM for ‘5-Series’ and ‘10-Series’, respectively, and $V_s = 5$ LPM through each of the reactors in ‘5-Parallel’ and ‘10-Parallel’ designs.

Series', '10-Series', '5-Parallel' and '10-Parallel', respectively. The numerical results of these configurations are compared with that of a single reactor at the same conditions. Here, the 5 kg LaNi₅ reactor discussed above is termed as 'Single Reactor'.

Fig. 7 shows the performance in terms of the absorption curves and the corresponding bed temperature profiles from simulations. The '5-Series', '10-Series', '5-Parallel' and '10-Parallel' designs show worse performance than 'Single Reactor' for the given set of conditions of P_s , V_s , and T_s (Fig. 7a and b). For example, at P_s : 15 bar, V_s : 5 LPM and T_s : 30 °C, the '5-Series' and '10-Series' designs take ~12% and ~26% longer times, respectively, than the 'Single Reactor' for 90% absorption (~1.26 wt%) as seen in Fig. 7a. Similarly, '5-Parallel' and '10-Parallel' designs take ~29% and ~47% longer times than the 'Single Reactor'. These trends are also reflected in the change in the average bed temperature with time (Fig. 7b). When the reactors are connected in series through the water line, the water entering the first reactor is at a temperature of T_s . However, as it enters the subsequent reactors, its temperature is higher than T_s . This is because it absorbs the heat released by the metal hydride during hydrogen uptake within the reactor it is exiting from. Hence, the cooling potential of water decreases when it enters the subsequent reactors. On the other hand, when the reactors are connected in parallel, the water supply splits equally into one-fifth or one-tenth in each of the reactors in '5-Parallel' and '10-Parallel' designs, respectively. This leads to slower rate of hydrogen absorption.

To enhance the performance of the scaled-up designs and match it with that of the 'Single Reactor', the parameters: P_s , V_s , and T_s can be changed. Among P_s , V_s , and T_s , it is easier to change V_s (i.e. water flow rate). Further, in our previous study we demonstrated that the water flow rate can have a great influence on the performance of such modular system [50]. Hence, we investigate on the performance of the '5-Series' and '10-Series' designs at higher water flow rates than 5 LPM. Fig. 7c and d shows the absorption curves and temperature profiles, respectively, of these scaled-up designs and compare with that of the 'Single Reactor' at 5 LPM. As seen from Fig. 7c and d, the volume flow rates of '5-Series' and '10-Series' designs need to be doubled ($V_s = 10$ LPM) and tripled ($V_s = 15$ LPM), respectively, to match the performance of the 'Single Reactor' at 5 LPM. An inherent assumption here is that the water does not lose any heat between its exit from a reactor and entry to the next one. On the other hand, for matching the performance of the reactors connected in parallel, each of the reactors needs to be supplied with 5 LPM of water. This amounts to a total of 25 and 50 LPM water flow rates in '5-Parallel' and '10-Parallel' designs, respectively. These enhancements in the performance of the '5-Parallel' and '10-Parallel' designs can be seen in Fig. 7c and d. Unlike the series configuration, the parallel configuration can have an advantage of being supplied with water at the same temperature in all the reactors. Further, the parallel configuration has advantage of less head loss of water flow than that in the series configuration. The present strategy helps in scaling-up a hydrogen storage system by exploiting the modular approach while retaining the advantages offered by the latter.

Table 1 – Comparison of the performance between the present reactor and those from literature.

Reference	LaNi ₅ Mass, kg	LaNi ₅ -bed Diameter, mm	LaNi ₅ -bed Height, mm	Heat transfer system used in literature	Time taken for 90% absorption, min	
					From Reference, (operating conditions)	Present Reactor (with same P_s , T_s . V_s increased proportionally for 5 kg of LaNi ₅)
Present Reactor	5	78	340	12 conical copper fins + 4 S tubes	–	–
Lewis et al. [57]	0.684	60	80	Embossed Plate Heat Exchanger	32 ($P_s = 10$ bar $T_s = 20$ °C Water flow 0.4639 g s ⁻¹)	31
Singh et al. [45]	1	52	140	13 copper perforated fins + ss central cooling tube	15 ($P_s = 15$ bar $T_s = 30$ °C $v_s = 1$ m s ⁻¹)	15
Lewis et al. [46]	0.390	60	47.5	4 U-tube with 3 fins each	11 ($P_s = 10$ bar $T_s = 20$ °C $v_s = 0.3$ m s ⁻¹)	13
Keshari et al. [48]	1			24 copper tubes + 13 sets of 12 copper pin fins	6.3 ($P_s = 15$ bar $T_s = 30$ °C $V_s = 6.75$ LPM)	11.25
Karmakar et al. [58]	10	114.3	376	4 cooling tubes + water jacket	20 ($P_s = 20$ bar $T_s = 30$ °C $V_s = 20$ LPM)	10

Comparison of the performance between the present reactor and those from literature

The performance of the reactor in the present study is compared with those from literature using LaNi_5 . For such a comparison, it is essential to keep the total mass of LaNi_5 and the operating conditions as the same between the present study and those from literature. Since various masses of LaNi_5 were employed in literature, the exact comparison of the performances is difficult. However, in order to facilitate a comparison with reactors smaller than the one in the present study [45,46,48,57], 5 kg of LaNi_5 is taken as the basis. For comparison with the reactor larger than the one in the present study [58], 10 kg of LaNi_5 is taken as the basis. Hence, for this case, the number of reactors from the present study is increased proportionally and considered to be connected in parallel. The parallel connection ensures that the duration for 90% absorption from these proportionally scaled-up configurations either from literature or the present study remains exactly the same as that for the corresponding smaller reactors. For obtaining the same operating conditions as used in literature, the volume flow rates (V_s) of water have to be proportionally increased or decreased prior to getting split before entering each of the reactors connected in parallel. This proportionally increased or decreased V_s is used in the 5 kg reactor of the present study for comparison. The pressure (P_s) and temperature (T_s) are chosen to be the same as those used in literature.

The comparison between the results obtained from the present study, considering the P_s , T_s , and the proportionally scaled-up V_s , and those reported in literature is given in Table 1. In most of the cases the results from the present study are either better or match closely with those from literature. For example, the reactor in Ref. [57] has only water jacket without any internal fins. However, the design of the present reactor utilizes internal fins and cooling tubes. Thus the present reactor, despite being much larger in size, can cool a much higher amount of LaNi_5 to reach 90% saturation at almost the same time. The internal conical copper fins of the present design perform equally when compared to a 1 kg LaNi_5 reactor featuring flat circular fins employed in Ref. [45]. The performance of the present reactor is almost the same as that reported by Lewis et al. [46]. Normally, a single reactor is expected to show a deteriorated performance than several smaller reactors operating together for the same operating conditions. The design proposal (simulation) by Keshari et al. [48] suggests better performance than the present reactor because of the presence of 24 cooling tubes (versus 4 in the present reactor) distributed evenly throughout the smaller 1 kg (1/5th of the current) LaNi_5 bed. However, fabricating such a reactor can be difficult. The present reactor performs better than the one in Ref. [58], due to the presence of the fins and the smaller reactor diameter. As seen from this comparison, the HTS design plays a crucial role in governing the performance, as systematically demonstrated in our earlier work [50]. Therefore, a careful choice of various design aspects such as MH-bed thickness, fin/MH-bed and fin/tube contact surface areas [50], combined with the present modular strategy and

the ease of fabrication can be used to enhance the practical hydrogen sorption performance of a reactor.

Conclusions

Hydrogen absorption and desorption performances of a 5 kg LaNi_5 reactor (SS 316, Sch- 40s) featuring novel conical copper fins connected by SS 316 heat transfer tubes is studied experimentally. Water is used as the heat transfer fluid. The performance of the reactor was evaluated at various conditions of water flow rate (V_s), hydrogen supply pressure (P_s), and water inlet temperature (T_s). As expected, higher V_s , P_s , and lower T_s result in faster absorption. These values of V_s , P_s and T_s offer higher driving force during absorption ($\propto \ln(P_s/P_{eq})$, P_{eq} : equilibrium pressure for absorption). For example, by increasing P_s from 10 to 25 bar the time taken for 90% absorption (~63 g) decreased from ~16.7 min to ~9.1 min (V_s : 5 LPM, T_s : 30 °C). Similarly, faster desorption is achieved by higher bed temperature and to some extent V_s , which offer higher driving force ($\propto (1-P_d/P_{eq})$, P_d : desorption pressure, ~1 atm). For example, at T_s : 60 °C, 90% desorption completed in ~20.1 min for T_i : 60 °C; and ~21.8 min for T_i : 30 °C at V_s : 5 LPM. Also, the desorption time decreased from ~22.3 min for V_s : 2.5 LPM to ~21.1 min for V_s : 7 LPM at T_i : 30 °C and T_s : 60 °C.

The absorption performance of this 'Single Reactor' is also numerically modeled and validated. Eventually, the numerical models of the scaled-up versions of five (25 kg LaNi_5) and ten (50 kg LaNi_5) such single reactor modules connected in series and parallel with water lines are also considered. These reactors are termed as '5-Series', '10-Series', '5-Parallel' and '10-Parallel', respectively, and their performance is compared with that of 'Single Reactor'. To match the performance of 'Single Reactor' at 5 LPM, the volume flow rates of '5-Series', '10-Series' configurations need to be just doubled (V_s = 10 LPM) and tripled (V_s = 15 LPM), respectively. Whereas, in the '5-Parallel' and '10-Parallel' configurations each of the reactors needs to be supplied with 5 LPM, thus necessitating the proportional increase in the water supply. However, the parallel configuration can be supplied with water at the same temperature to all the reactors. The present work shows the comparison of the performances of series and parallel configurations.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijhydene.2022.08.098>.

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